Rajneesh Anand

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Research Interests

Optimal Control, Reinforcement Learning, Microchemical Systems, Computational Fluid Dynamics, Transport Processes, and Soft Matter

Education

Lehigh University, Bethlehem, PA, USA — Aug, 2024 – Present
PhD, Chemical Engineering — Current GPA: 3.97/4.0

National Institute of Technology (NIT) Durgapur, WB, India — 2019 – 2023
Bachelor of Technology (B.Tech), Chemical Engineering — GPA: 8.30/10.0 (US: 3.60/4.0)
First Class with Distinction

Academic Experience

Research Assistant — Oct, 2024 – Present
Department of Chemical and Biological Engineering, Lehigh University, Bethlehem, PA, USA

- Advisor: Dr. Mayuresh V. Kothare
- *Currently working on the intersection of reinforcement learning, optimal control, and microfluidics*

Research Assistant — May, 2022 – July, 2023
Department of Chemical Engineering, National Institute of Technology (NIT) Durgapur, WB, IN

- Advisors: Dr. Abhiram Hens and Dr. Mrinal Kanti Mandal
- Thesis: *Optimisation of geometry and surface wettability for enhanced efficiency of the solar still-based desalination unit*
- Project sponsored by the Dept. of Science & Tech. (India)

Industrial Experience

Reliance Industries Ltd., Jamnagar, India — Aug – Dec, 2023
Graduate Engineer Trainee (Process Engineer)

Mitsubishi Chemical Corporation, Haldia, WB, India — May – June, 2022
Industrial Engineering Intern

Honors and Awards

Third Prize (National Level) in the **New Generation Ideation Contest 2022** (NGIC), issued by HP Green R&D Centre (HPCL, India). *Topic: Bi-layered ceramic membrane single separator-photo-reactor for the capture & conversion of CO_2 to CH_4.* — 2023

Mentoring Experience

Lehigh University, Bethlehem, PA

Fatima Tanveer (Lafayette College, Undergraduate) – co-mentored with M.V. Kothare — Summer, 2026
Supervised DropletRunner project. Won 'Best Research' award in Lehigh's Summer Research Expo 2026.

National Institute of Technology (NIT) Durgapur, WB, India

Meghna Bhattacharya (NIT Durgapur, Undergraduate) — Summer, 2023
Supervised CFD-based research on heat exchangers; work presented in IIChE-CHEMCON conference **[C6]**.

Teaching Experience

Lehigh University, Bethlehem, PA

Fluid Mechanics, Teaching Assistant — Spring, 2026

Heat and Mass Transfer, Teaching Assistant — Fall, 2025

Journal Publications

[J4] **Anand, R.**, Kothare, M.V.* Optimal control and closed-loop stability of droplet transport in a microchannel. *(Manuscript under revision)*

[J3] **Anand, R.**, Kothare, M.V.* (2026) Surfactant-free splitting of water droplets in a microchannel. ***Analyst***.

[J2] **Anand, R.**, Mandal, M.K., Chaudhuri, R.G., Hens, A.* (2024). Effect of patterned condensation surface on the performance of a newly designed basin-type solar desalination unit. ***Applied Thermal Engineering***.

[J1] **Anand, R.**, Mandal, M.K., Chaudhuri, R.G., Hens, A.* (2023). Optimization of geometry and surface wettability for enhanced efficiency of the solar still-based desalination unit. ***Int. Journal of Ambient Energy***.

Book Chapters

[B1] Akhtar, F., **Anand, R.*** (2025). Computational study of flow past a cylinder in tandem setup. ***Advances in Mechanical Engineering*** (Springer Nature).

Refereed Abstracts

[A10] **Anand, R.**, Kothare, M.V.* (2026). Autonomous Droplet Navigation In a Labyrinth Via Reinforcement Learning. ***AIChE** Annual meeting 2026*.

[A9] **Anand, R.**, Kothare, M.V.* (2026). Optimal Control and Closed-Loop Stability of Droplet Transport In a Microchannel. ***AIChE** Annual meeting 2026*.

[A8] **Anand, R.**, Kothare, M.V.* (2026). Surfactant-free droplet breakup dynamics under confinement. ***APS** Global Physics Summit (March Meeting) 2026*.

[A7] **Anand, R.**, Kumar, I., Hens, A.* (2023). Effect of bed porosity and inlet temperature on hydrogen adsorption inside a cylindrical storage tank: A CFD study. ***IIChE-CHEMCON** 2023*.

[A6] Bhattacharya, M., Akhtar, F., **Anand, R.*** (2023). Numerical investigation of heat transfer enhancement in heat exchangers using CFD. ***IIChE-CHEMCON** 2023*.

[A5] **Anand, R.***, Ray, R., Akhtar, F. (2023). Modeling and simulation of cumene process in Aspen Plus. *Int. Conf. on Recent Advances in Technology, Engineering, Science and Management (**ICRATESM**-2023)*.

[A4] Akhtar, F., **Anand, R.**, Hens, A.* (2023). Computational study of pipeline transportation of crude oil. *Int. Conference on Chemical Engineering Innovations and Sustainability (**ICEIS**-2023), Jadavpur University*.

[A3] Akhtar, F., **Anand, R.*** (2023). Computational study of flow past a cylinder in tandem setup. *4^{th} Int. Conference on Recent Advancements in Mechanical Engineering (**ICRAME**-2023)*.

[A2] Akhtar, F., & **Anand, R.*** (2023). The adsorptive hydrogen storage in an activated carbon tank based on a modified Dubinin–Astakhov model. *Int. Conference on Materials Science and Mechanical Engineering (**ICMSME**-2023)*.

[A1] **Anand, R.***, Akhtar, F. (2022). Effect of operating parameters on nickel removal in a fixed bed adsorption column. *Int. Conference on Advances and Creations in Mechanical Engineering (**ICACME**)*.

* denotes corresponding author.

Services

Peer Review
Journals: IEEE Control Systems Letters, Automatica
Conferences: Conference on Decision and Control

Software Products

[S2] **DropletRunner** (Main Developer)

- Autonomous droplet navigation system via MBRL on open surfaces
- Simulation-free pipeline integrating ROS2 vision, Dynamixel actuation, and DreamerV3
- https://github.com/rajneeshanand/DropletRunner

[S1] **OncoLocus** (Main Developer)

- Hierarchical Temporal Transformer for Cancer NLP
- Cross-cancer zero-shot transfer to three unseen cancer types
- https://github.com/rajneeshanand/OncoLocus

Technical Skills

Robotics & Systems:	ROS2, Circuit Design, Distributed Computing, Computer Vision, & CAD
Programming:	MATLAB, Python, Parallel Computing & HPC
Numerical Methods:	Continuous (Nonlinear) Optimization, FEM, FVM, PDE Solvers
CFD Software:	ANSYS Fluent, COMSOL Multiphysics, FEniCS
Process Simulation:	Aspen Plus

References

Provided upon request.